

① BCS Driven system:

$$\dot{p} = -i[H_0, p] + \gamma_0 D[p]$$

$$U^{MC}(t) = \sum_k' (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k & -\Delta \\ -\Delta & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \xi_k$$

$$= (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k c_k + \Delta c_{-k}^\dagger \\ \Delta c_k - \xi_k c_{-k}^\dagger \end{pmatrix} =$$

$$= c_k^\dagger c_k \xi_k + \Delta c_k^\dagger c_{-k}^\dagger + \Delta c_{-k} c_k - \xi_k c_k c_{-k}^\dagger =$$

$$= \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) + \Delta (c_k^\dagger c_{-k}^\dagger + c_{-k} c_k) - \xi_k c_k c_{-k}^\dagger$$

Bozonsymmetrie: $\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} m & r \\ -r & m \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}$

↳ gleich notw. x opp.

gap: $\Delta_k(t) = \sum_{nn'} V_{nn'} \text{Tr}[c_k^\dagger c_{k'} p(t)]$

$$\rightarrow E_k(t) = \pm \sqrt{\xi_k^2 + \Delta_k^2(t)} \quad \text{mit} \quad M_k(t) = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k(t)} \right), \quad N_k(t) = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k(t)} \right)$$

→ simple modellierung kopuli:

$$D[\cdot] = D^{(T_1)}[\cdot] + D^{(T_2)}[\cdot]$$

$$D^{(T_1)} p = \sum_{\substack{\text{r.f.f.} \\ k}} L_p L^\dagger - \frac{1}{2} \{p, L L^\dagger\}$$

$$L_{kp}^{(s,r)} = \sqrt{\gamma(t)} \alpha_k^{(s)} \alpha_p^{(r)}$$

$$\alpha^{(+)} = \alpha^\dagger$$

$$\alpha^{(-)} = \alpha$$

$$k = (k, \sigma)$$

$$p(t) = \frac{1}{2} \exp \left[\sum_k \mu_k(t) \alpha_k^\dagger \alpha_k \right]$$

$$\langle n_k \rangle = n_k = \langle \alpha_k^\dagger \alpha_k \rangle$$

then

$$\rightarrow \text{EOM: } \langle \dot{n}_k \rangle = \text{Tr}[n_k D p] = \gamma_0 \sum_j \left(\langle L_j^\dagger n_k L_j \rangle_p - \langle L_j^\dagger L_j n_k \rangle_p \right)$$

$$s_1 s_2 = \pm 1$$

$$\rightarrow \langle L_j^\dagger n_k L_j \rangle \propto \sum_{p \neq j} \langle \alpha_p^\dagger \alpha_p^\dagger \alpha_k^\dagger \alpha_k^\dagger \alpha_p \alpha_p \rangle = \text{6 terms.}$$

$$\langle \alpha_k^\dagger \alpha_p \rangle = n_k \delta_{ks}$$

EOM:

$$\langle \dot{n}_k \rangle = \text{Tr}[n_k(t) \dot{p}(t)] = -i \text{Tr}[n_k(t) [U(t), p(t)]] + \gamma_0 \text{Tr}[n_k(t) D p] =$$

$$= \gamma_0 \text{Tr}[n_k (D_1 p + D_2 p)] = \gamma_0 \text{Tr}[n_k D^{(T_1)} p] \quad \text{mit} \quad D_1 = D^{(T_1)} \quad \text{und} \quad L_1 = \sqrt{\gamma(t)} \alpha_k^\dagger \alpha_k \dots$$

$$= r_0 \text{Tr} \left(n_k \sum_{\substack{s_1 s_2 \\ 2p}} \gamma_{2p}^{s_1 s_2} (T_1) \alpha_p^{s_1} \alpha_p^{s_2} \left\{ \alpha_p^{s_1+} \alpha_p^{s_2+} - \frac{1}{2} \delta_{2p} \right\} \rho, \alpha_p^{s_1+} \alpha_p^{s_2+} \alpha_p^{s_1} \alpha_p^{s_2} \right) + r_0 (T_2) =$$

$$= r_0 \sum_{\substack{s_1 s_2 \\ 2p}} \gamma_{2p}^{s_1 s_2} (T_1) \left(\langle \alpha_p^{s_1+} \alpha_p^{s_2+} n_k \alpha_p^{s_1} \alpha_p^{s_2} \rangle - \langle \alpha_p^{s_1+} \alpha_p^{s_2+} \alpha_p^{s_1} \alpha_p^{s_2} n_k \rangle \right) + r_0 (T_2) =$$

$$\text{Tr} [f \frac{1}{2} \{p, A\}] = \text{Tr} \left[\frac{1}{2} f p A + \frac{1}{2} f A p \right] = \frac{1}{2} \text{Tr} [f p A + f A p]$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \cdot p] = \frac{1}{2} \langle A f + f A \rangle$$

if $f = n_k \rightarrow \langle A f \rangle$

→ compare: $s_1 s_2 = +1$

$$(\dagger) \langle \alpha_p \alpha_z \alpha_k^+ \alpha_k^+ \alpha_z^+ \alpha_p^+ \rangle - \langle \alpha_p \alpha_z \alpha_z^+ \alpha_p^+ \alpha_k^+ \alpha_k^+ \rangle = \langle [n_k, \alpha_p^+] \rangle = \langle \alpha_k^+ \alpha_k, \alpha_p^+ \rangle =$$

$$= \langle \alpha_p \alpha_z [n_k, \alpha_z^+] \rangle =$$

$$[n_k, \alpha_z^+] = [n_k, \alpha_z^+] \alpha_p^+ + \alpha_z^+ [n_k, \alpha_p^+] = \alpha_k^+ \alpha_p^+ \delta_{kz} + \alpha_z^+ \alpha_k^+ \delta_{kp}$$

$$= \langle \alpha_p \alpha_z (\alpha_k^+ \alpha_p^+ \delta_{kz} + \alpha_z^+ \alpha_k^+ \delta_{kp}) \rangle = (\delta_{kz} + \delta_{kp}) \langle \alpha_p \alpha_z \alpha_z^+ \alpha_p^+ \rangle =$$

$$\alpha \alpha^+ = 1 - \alpha^+ \alpha = 1 - n$$

$$= (\delta_{kz} + \delta_{kp}) \cdot (1 - \delta_{2p}) \langle \alpha_p \alpha_p^+ \alpha_z \alpha_z^+ \rangle =$$

$$= (\delta_{kz} + \delta_{kp}) (1 - \delta_{2p}) (1 - n_p) (1 - n_z)$$

$$\boxed{T_{2p} = T_{12}}$$

$$\dot{n}_k = r_0 \sum_{2p} \Gamma_{2p}^{++} (\delta_{kz} + \delta_{kp}) (1 - \delta_{2p}) (1 - n_p) (1 - n_z) + 3 \text{ term} =$$

$$= r_0 \left(\sum_p \left[\Gamma_{kp}^{++} (1 - \delta_{kp}) (1 - n_p) (1 - n_k) \right] + \sum_{\substack{z \\ 2 \rightarrow p}} \left[\Gamma_{kz}^{++} (1 - \delta_{kz}) (1 - n_k) (1 - n_z) \right] \right) =$$

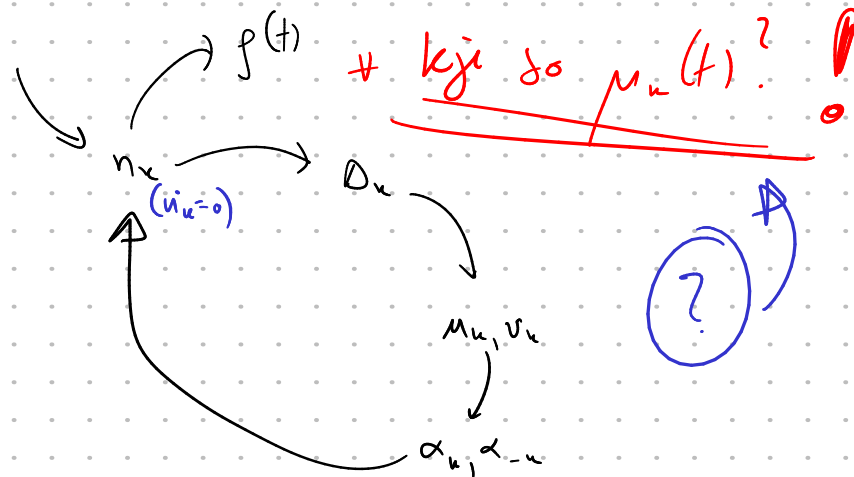
$$= r_0 \cdot 2 \sum_p \Gamma_{kp}^{++} (1 - \delta_{kp}) (1 - n_p) (1 - n_k) + 3 \text{ term}$$

$$2 \sum \Gamma (1-n)(1-n) - r_0 2 \Gamma_{kk}^{++} (1-n_k)^2 = \dots - 2 r_0 \Gamma_{kk}^{++} n_k$$

prima δ_1 ? 0.1 ?

skupaj: ① enačba za $\dot{n}_k(t) = f(n_k, \dots)$

② gup: $\Delta_k(t) = \sum_k \frac{V_{kk'}}{2} \frac{1}{\sqrt{\xi_k^2 + \Delta_k^2(t)}} \cdot (1 - n_k(t) - n_{-k}(t))$



skupaj:

(+)

$$\sum_{2P} \Gamma_{2P}^{+-} (\delta_{k2} - \delta_{kP}) n_P (1 - n_2) = \sum_P (\Gamma_{kP}^{+-} n_P (1 - n_k) - \Gamma_{P2}^{+-} n_k (1 - n_P)) \quad (+)$$

(-+) $\Gamma_{2P}^{+-} = \Gamma_{P2}^{+-}$ (?) preveri!

$$-\sum_{2P} \Gamma_{2P}^{+-} (\delta_{k2} - \delta_{kP}) n_2 (1 - n_P) = \sum_P \Gamma_{Pk}^{+-} (-n_k (1 - n_P)) + \sum_P \Gamma_{kP}^{+-} n_P (1 - n_k) = \sum_P (\Gamma_{kP}^{+-} n_P (1 - n_k) - \Gamma_{Pk}^{+-} n_k (1 - n_P)) \quad (-)$$

samo en
člen!

$$\dot{n}_k = 0 = 2 \sum_j \left(\Gamma_{kp}^{++} (1-n_k)(1-n_p) - \Gamma_{kp}^{--} n_k n_p \right) + \sum_p \left(\Gamma_{kp}^{+-} (n_p - n_k) - \Gamma_{pk}^{+} (n_k - n_p) \right) = \quad ?$$

$$\alpha = \frac{\alpha_{++}}{\alpha_{++} + \alpha_{--}} \quad , \quad \gamma_{s_1 s_2}^{s_1 s_2} = \alpha_{s_1 s_2} f_{s_1} \left(\frac{\epsilon_k}{T} \right) f_{s_2} \left(\frac{\epsilon_p}{T} \right)$$

$$\begin{aligned} \rightarrow & \quad 2 \alpha_{++} \left(f_{+} \left(\frac{\epsilon_k}{T_1} \right) f_{+} \left(\frac{\epsilon_p}{T_1} \right) + f_{+} \left(\frac{\epsilon_k}{T_2} \right) f_{+} \left(\frac{\epsilon_p}{T_2} \right) \right) (1-n_k)(1-n_p) - \\ & - 2 \alpha_{--} \left(f_{-} \left(\frac{\epsilon_k}{T_1} \right) f_{-} \left(\frac{\epsilon_p}{T_1} \right) + f_{-} \left(\frac{\epsilon_k}{T_2} \right) f_{-} \left(\frac{\epsilon_p}{T_2} \right) \right) n_k n_p + \\ & + \alpha_{+-} \left[\left(f_{+} \left(\frac{\epsilon_k}{T_1} \right) f_{-} \left(\frac{\epsilon_p}{T_1} \right) + f_{-} \left(\frac{\epsilon_k}{T_2} \right) f_{+} \left(\frac{\epsilon_p}{T_2} \right) \right) (1-n_k)n_p - \right. \\ & \left. - \left(f_{+} \left(\frac{\epsilon_p}{T_1} \right) f_{-} \left(\frac{\epsilon_k}{T_1} \right) + f_{+} \left(\frac{\epsilon_p}{T_2} \right) f_{-} \left(\frac{\epsilon_k}{T_2} \right) \right) n_k(1-n_p) \right] = \quad \text{resumo} \quad \text{(kind of)} \quad \text{???} \end{aligned}$$

$$\begin{aligned} = & \sum_{T_j} \left[\sum_p \left(f_{+} \left(\frac{\epsilon_p}{T_j} \right) (1-n_p) + \alpha f_{-} \left(\frac{\epsilon_p}{T_j} \right) n_p \right) \cdot (1-n_k) f_{+} \left(\frac{\epsilon_k}{T_j} \right) - \right. \\ & \left. - \left(\sum_p \left(f_{-} \left(\frac{\epsilon_p}{T_j} \right) n_p + \alpha f_{+} \left(\frac{\epsilon_p}{T_j} \right) (1-n_p) \right) n_k f_{-} \left(\frac{\epsilon_k}{T_j} \right) \right] = \dot{n}_k = \end{aligned}$$

$$= \sum_{T_j} \left(A(T_j) (1-n_k) f_{+} \left(\frac{\epsilon_k}{T_j} \right) - B(T_j) n_k f_{-} \left(\frac{\epsilon_k}{T_j} \right) \right) = \dot{n}_k = 0$$

Fokscitandhi itobatangzi + thermal baths :

$$H = \sum_k (a_k^\dagger b_k^\dagger) \begin{pmatrix} \epsilon_a + V_{ab}(t) & \Delta_k^*(t) \\ \Delta_k(t) & \epsilon_b + V_{ba}(t) \end{pmatrix} \begin{pmatrix} a_k(t) \\ b_k(t) \end{pmatrix} \rightarrow \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} = \begin{pmatrix} u & v \\ -v^* & u \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

$$\Delta_k(t) = - \sum_{k'} V_{kk'} \text{Tr}[a_{k'}^\dagger b_{k'} \rho(t)]$$

$$\bullet E_{\pm}(t) = \frac{\epsilon_a(t) + \epsilon_b(t)}{2} \pm \sqrt{\left(\frac{\epsilon_a - \epsilon_b}{2}\right)^2 + \Delta_k^2} = \eta_k(t) \pm \sqrt{\xi_k^2(t) + \Delta_k^2(t)}$$

$$\Delta_k(t) = - \sum_{k'} \frac{V_{kk'}}{2} \cdot \frac{1}{\sqrt{\xi_k^2(t) + \Delta_k^2(t)}} \cdot (\langle \alpha_k^\dagger \alpha_k \rangle - \langle \beta_k^\dagger \beta_k \rangle)$$

$$\bullet M_k^1 = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \right), \quad M_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \right)$$